

1 Series of functions

- **Series of functions** are numerical series depending on a parameter x : when these series converge for every value x belonging to a certain interval, they represent **functions of a new kind**
- There are two important classes of series of functions, namely trigonometric series and power series. We will only deal with **power series**, also called **Taylor series**, which can be seen as a natural extension of Taylor expansions.

1.1 Taylor series of the elementary transcendental functions

- Taylor's expansion/formula with Lagrange remainder can be rewritten in the form:

$$f(x) = \sum_{k=0}^n \frac{f^{(k)}(x_0)}{k!} (x - x_0)^k + E_n(x) \quad (1)$$

where the Lagrange approximation error is

$$E_n(x) = \frac{f^{(n+1)}(c)}{(n+1)!} (x - x_0)^{n+1} \quad (2)$$

and c is a suitable point between x_0 and x .

Definition 1: Taylor series

If a function f has derivatives of every order, the series

$$\sum_{k=0}^{\infty} \frac{f^{(k)}(x_0)}{k!} (x - x_0)^k$$

is called the **Taylor series** of the function f centered at x_0 .

At a given point $x \in (a, b)$, if the error

$$E_n(x) \rightarrow 0 \text{ as } n \rightarrow +\infty$$

then the Taylor series is convergent and its sum equals $f(x)$. This is equivalent to saying that the n -th partial sum of the series has a finite limit and this limit is precisely $f(x)$. In formulas:

$$\underbrace{T_{n,x_0}(x)}_{\text{Taylor polynomial}} = \sum_{k=0}^n \frac{f^{(k)}(x_0)}{k!} (x - x_0)^k \rightarrow f(x) \text{ as } n \rightarrow +\infty$$

Definition 2: function expandable in Taylor series on an interval

Given a function $f : D \rightarrow \mathbb{R}$ with derivatives of every order, if

$$E_n(x) \rightarrow 0 \text{ as } n \rightarrow +\infty, \quad \forall x \in I \subseteq D$$

we say that $f(x)$ is **expandable in Taylor series** on the interval I .

- There are (infinitely differentiable) functions that are expandable in Taylor series on the whole real line, others that are expandable only on a bounded interval, and others for which the interval reduces to a single point (the point x_0).

- We now deal with the three elementary transcendental functions:

$$f(x) = e^x, \quad f(x) = \sin x \quad \text{and} \quad f(x) = \cos x$$

which we will show to be expandable in Taylor series on the whole of $\mathbb{R}$. They are three transcendental (not algebraic) functions.

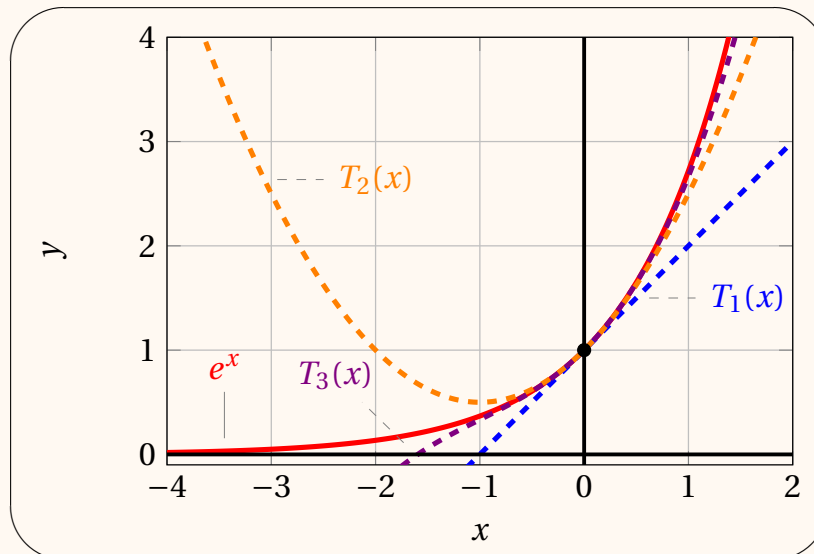
- Real functions of a real variable are classified into **algebraic functions and transcendental functions**. Algebraic functions are built through a finite number of applications of the four arithmetic operations, of raising to a power and of extracting the n -th root. Transcendental functions are all the functions that are not algebraic.

1.1.1 The Taylor series of the exponential function

Example 1: Maclaurin formula/expansion of the exponential

Maclaurin formula/expansion of order n of the exponential with Lagrange remainder:

$$e^x = T_n(x) + \frac{e^c}{(n+1)!} x^{n+1} = 1 + x + \frac{x^2}{2} + \frac{x^3}{3!} + \dots + \frac{x^n}{n!} + \frac{e^c}{(n+1)!} x^{n+1} \quad \text{with } c \in [0, x]$$



Remark 1: Taylor series of the exponential function

$$\forall x \in \mathbb{R}, \quad e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!}$$

Proof. From the Maclaurin expansion of order n of e^x with Lagrange remainder, we have that for every integer n and $x \in \mathbb{R}$ there exists a point c , lying between 0 and x , such that:

$$e^x = \sum_{k=0}^n \frac{x^k}{k!} + \frac{e^c}{(n+1)!} x^{n+1}$$

We fix x and let n tend to $+\infty$. The point c may vary with n but, since it always lies between 0 and x , we have:

$$e^c \leq \begin{cases} e^x & \text{if } x > 0 \\ 1 & \text{if } x < 0 \end{cases}$$

and $e^0 = 1$, hence e^c stays bounded. From the hierarchy of infinities theorem we have:

$$\frac{x^{n+1}}{(n+1)!} \rightarrow 0 \text{ as } n \rightarrow +\infty$$

hence the error tends to zero, that is:

$$\frac{x^{n+1}}{(n+1)!} e^c \rightarrow 0 \text{ as } n \rightarrow +\infty$$

since it is the product of an infinitesimal sequence and a bounded one. We have therefore proved that the function e^x can be written as the sum of a power series, its Taylor series, convergent for every $x \in \mathbb{R}$. $\square$

Remark 2

$$e = \sum_{k=0}^{\infty} \frac{1}{k!} \tag{3}$$

Proof. It suffices to set x equal to 1 in the Taylor series of the exponential function. $\square$

- We have derived a second definition of Euler's number e (Napier's constant); the first one was:

$$e = \lim_{k \rightarrow +\infty} \left(1 + \frac{1}{k}\right)^k \tag{4}$$

- Consequently, we have two methods for computing an approximation of the value of e :

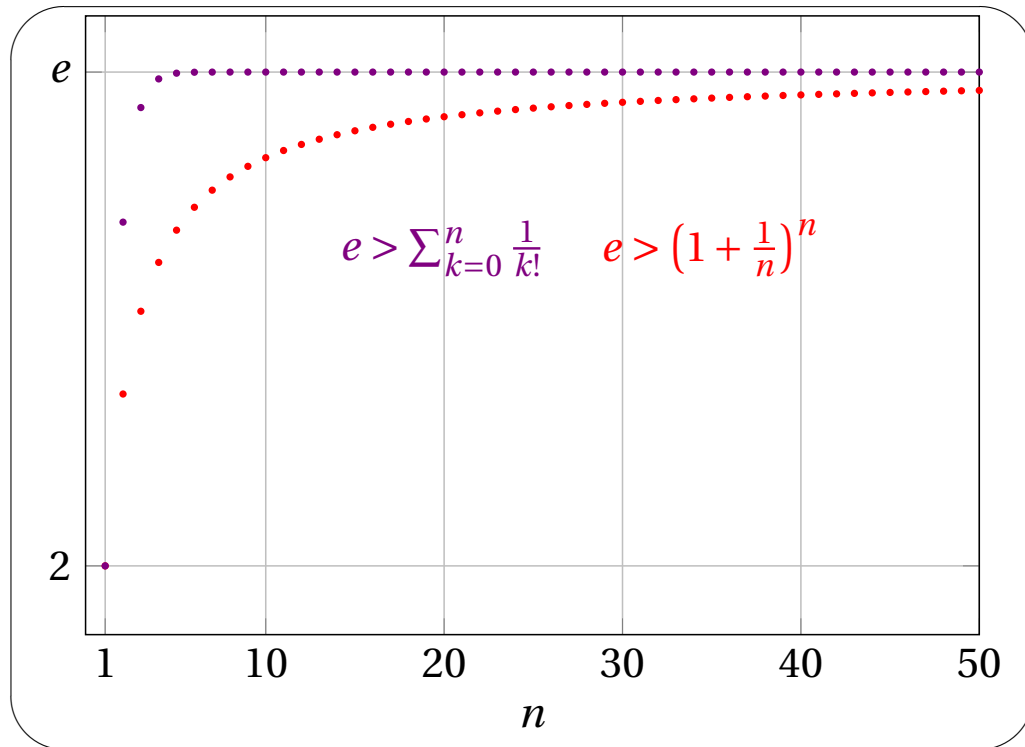
- The first method is based on formula (4), and by fixing $k = n$ we obtain the approximation:

$$e \approx \left(1 + \frac{1}{n}\right)^n$$

- The second method is based on formula (3), and by computing the n -th partial sum we obtain the approximation:

$$e \approx \sum_{k=0}^n \frac{1}{k!}$$

	$\left(1 + \frac{1}{n}\right)^n$	$\sum_{k=0}^n \frac{1}{k!}$	e
$n = 1$	2.0000000000...	2.0000000000...	2,7182818284...
$n = 2$	2.2500000000...	2.5000000000...	2,7182818284...
$n = 3$	2.3703703704...	2.6666666666...	2,7182818284...
$n = 4$	2.4414062500...	2.7083333333...	2,7182818284...
$n = 5$	2.4883200000...	2.7166666666...	2,7182818284...



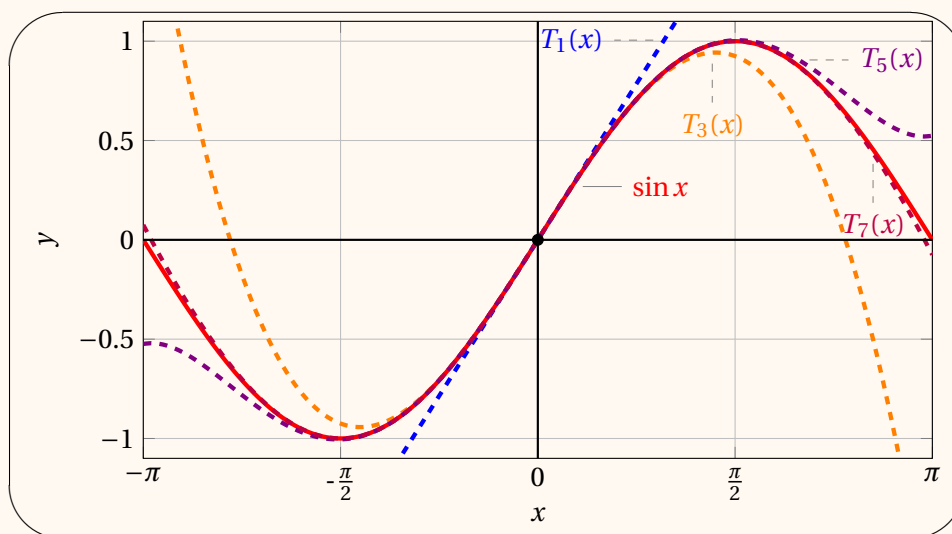
- The second method is therefore much more efficient at approximating the value of e

1.1.2 The Taylor series of the elementary trigonometric functions

Example 2: Maclaurin formula/expansion of sine and cosine

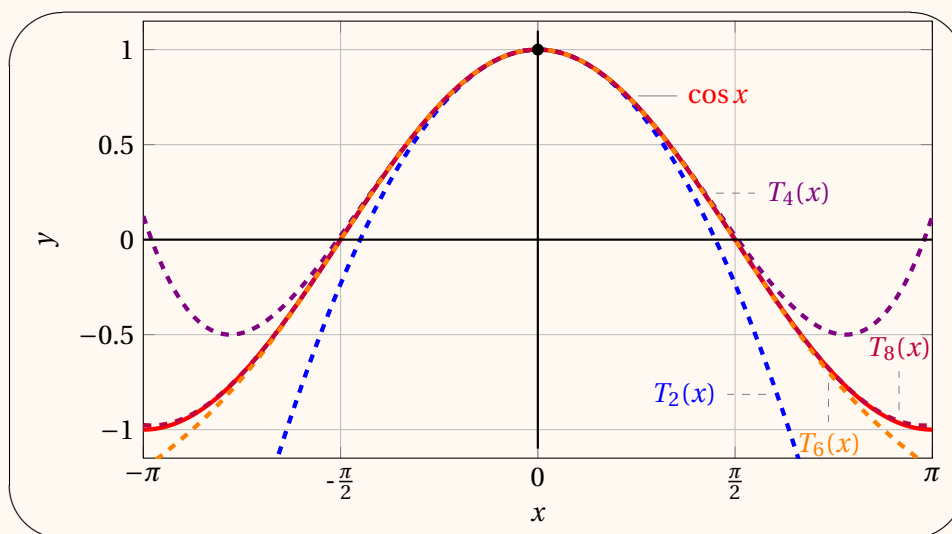
Maclaurin formula/expansion of odd order of the sine with Lagrange remainder:

$$\begin{aligned}\sin x &= T_{2k+1}(x) + \frac{\sin^{(2k+2)}(c)}{(2k+2)!} x^{2k+2} \\ &= x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots + (-1)^k \frac{x^{2k+1}}{(2k+1)!} + \frac{\sin^{(2k+2)}(c)}{(2k+2)!} x^{2k+2} \text{ with } c \in [0, x]\end{aligned}$$



Maclaurin formula/expansion of even order of the cosine with Lagrange remainder:

$$\begin{aligned}\cos x &= T_{2k}(x) + \frac{\cos^{(2k+1)}(c)}{(2k+1)!} x^{2k+1} \\ &= 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots + (-1)^k \frac{x^{2k}}{(2k)!} + \frac{\cos^{(2k+1)}(c)}{(2k+1)!} x^{2k+1} \text{ with } c \in [0, x]\end{aligned}$$



Remark 3: Taylor series of the elementary trigonometric functions

$$\forall x \in \mathbb{R}, \quad \sin x = \sum_{k=0}^{\infty} (-1)^k \frac{x^{2k+1}}{(2k+1)!}, \quad \cos x = \sum_{k=0}^{\infty} (-1)^k \frac{x^{2k}}{(2k)!}$$

Proof.

□

2 Series in the complex field